THE UNITED AFRICAN UNIVERSITY OF TANZANIA

COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES

PRACTICAL TRAINING LOG-BOOK & TECHNICAL COMPENDIUM

Host Institution: Tanzania Telecommunication Corporation (TTCL) • Academic Year: 2025/2026

Field Specialization: Information Systems, Networking & Software Development

STUDENT'S NAME: DAVID ERICK MWIJAGE **COMPANY / INSTITUTION:** TANZANIA TELECOMMUNICATION CORPORATION (TTCL)	**REG. NO:** 25013055 **ACADEMIC YEAR:** 2025/2026 (BSc ICT)

WEEK NO: 1 | FROM: 24th August 2026 TO: 28th August 2026

DAY / DATE	ACTIVITY / TASKS EXECUTED
Monday *24/08/2026*	**ORIENTATION & DISTRIBUTION OF DUTIES** Introduction to TTCL staff, corporate structure, operational procedures, code of conduct, and formal distribution of field training duties in the IT and network department.
Tuesday *25/08/2026*	**PUBLIC HOLIDAY** Official National Public Holiday. No industrial operational tasks scheduled.
Wednesday *26/08/2026*	**ENTERPRISE NETWORK LAYOUT & CABLING INFRASTRUCTURE** Studied building network layout, structured cabling, patch panels, distribution switches, access layers, and interconnection of telecommunication network devices across floors.
Thursday *27/08/2026*	**COMPUTER HARDWARE & PERIPHERALS MAINTENANCE** Learned foundational hardware maintenance procedures: internal component diagnostics, RAM testing, power supply units, network printers, and desktop peripheral troubleshooting.
Friday *28/08/2026*	**PROJECT KICKOFF: AVOCADO FARMERS SYSTEM IN PHP** Commenced major software development task: Assigned to engineer the Avocado Farmers Registration & Digital Marketplace System for Njombe Region using HTML, CSS, JavaScript, and native PHP with XAMPP stack.

DETAILS OF THE MAIN JOB OF THE WEEK

Operation: Hardware inspection, network cabling architecture tracing, and local PHP/MySQL project bootstrap.

Machinery / Tools Used: Desktop Workstations, Network Patch Panels, RJ45 Cat6 Cables, XAMPP Server (Apache, MariaDB), VS Code.

📸 WEEK 1 — PRACTICAL EVIDENCE & PHOTO GALLERY

Attach or paste printed field photographs, hardware inspections, and system screenshots in the designated boxes below:

[FIGURE 1.1: TTCL Building Network Switch & Rack Layout]

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Technical Caption & Observations (26/08/2026):
Inspection of floor distribution switch, structured UTP patch panel cabling, and trunk uplinks.

[FIGURE 1.2: Computer Hardware Maintenance & Diagnostics]

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Technical Caption & Observations (27/08/2026):
Preventative maintenance on office desktop units, motherboard cleaning, and peripheral testing.

[FIGURE 1.3: Avocado Farmers System Project Kickoff]

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Technical Caption & Observations (28/08/2026):
Initial system architecture setup in XAMPP htdocs, database schema draft, and login interface.

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WEEK NO: 2 | FROM: 31st August 2026 TO: 04th September 2026

DAY / DATE	ACTIVITY / TASKS EXECUTED
Monday *31/08/2026*	**COMPUTER MAINTENANCE & DISK OPTIMIZATION** Physical hardware de-dusting of slow desktop PCs, operating system updates, and automated disk cleanup routines. Purged temp caches and resolved slow booting bottlenecks, freeing 9.2 GB disk storage.
Tuesday *01/09/2026*	**SYSTEM DESIGN & FRONTEND USER INTERFACE BASICS** Refined user experience principles for smallholder farmers. Studied responsive CSS Grid and Flexbox layouts to ensure clean usability across desktop and mobile devices.
Wednesday *02/09/2026*	**AVOCADO FARMERS SYSTEM PRESENTATION & PHASE 1 REVIEW** Formally presented Phase 1 working prototype built in HTML, CSS, PHP to supervisors and stakeholders. Received valuable review feedback, architectural directives, and security enhancement tasks.
Thursday *03/09/2026*	**BACKEND REFACTORING & SECURE DATABASE CONNECTION** Refactored legacy database layer into modular architecture (config/db.php, .env). Enforced utf8mb4 encoding, silenced production errors to logs/php_errors.log, and set up prepared statements (mysqli_prepare).
Friday *04/09/2026*	**PRODUCT INVENTORY & STOCK MANAGEMENT MODULE** Engineered agronomic stock module: phenological growth tracking (growing, flowering, harvest-ready, harvested), tree count yield forecasting (kg), and smallholder floor-price threshold (min_price_per_kg).

DETAILS OF THE MAIN JOB OF THE WEEK

Operation: Preventative hardware maintenance, system design presentation, backend security refactoring, and inventory lifecycle modeling.

Machinery / Tools Used: Hardware Blower & Toolkit, Windows Disk Cleanup, XAMPP Server, MariaDB Database, VS Code, Browser DevTools.

WEEK 2 — PRACTICAL EVIDENCE & PHOTO GALLERY

Attach or paste printed field photographs, hardware inspections, and system screenshots in the designated boxes below:

[FIGURE 2.1: Desktop PC Maintenance & Disk Cleanup]

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Technical Caption & Observations (31/08/2026):
Execution of disk cleanup and hardware dust removal, resulting in 9.2 GB space recovery and faster boot times.

[FIGURE 2.2: Phase 1 Avocado System Stakeholder Presentation]

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Technical Caption & Observations (02/09/2026):
Demonstrating the initial web prototype to supervisors, gathering feedback on security and modular design.

[FIGURE 2.3: Backend Refactored Architecture & Inventory UI]

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Technical Caption & Observations (04/09/2026):
Implementation of decoupled config/db.php, environment variables, and produce inventory status cards.

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WEEK NO: 3 | FROM: 07th September 2026 TO: 11th September 2026

DAY / DATE	ACTIVITY / TASKS EXECUTED
Monday *07/09/2026*	**TTCL LAN DIAGNOSTICS & SWITCH PORT CONFIGURATION** Conducted local area network diagnostics at TTCL office. Learned IPv4 subnetting classes, verified switch port configurations, crimped RJ45 patch cords (T568B standard), and mapped device IP assignments using ping, traceroute, and arp commands.
Tuesday *08/09/2026*	**LINUX SERVER FUNDAMENTALS & APACHE SERVICE MANAGEMENT** Studied Linux server fundamentals using enterprise Unix documentation (Paul Cobbaut text). Practiced terminal CLI commands for directory navigation, file permissions (chmod, chown), process monitoring (top, ps aux), and managed web server daemon services (systemctl status apache2/mysql).
Wednesday *09/09/2026*	**ROLE-BASED ACCESS CONTROL (RBAC) & SESSION SECURITY** Implemented Role-Based Access Control (RBAC) in Avocado system for SuperAdmin, Admin, Agricultural Officer, Farmer, and Buyer roles (includes/auth.php). Added session timeout guard (15-minute idle limit) and automatic modal alert to terminate unauthorized sessions.
Thursday *10/09/2026*	**RELATIONAL DATABASE INDEXING & SQL QUERY OPTIMIZATION** Studied relational SQL principles from McGraw-Hill enterprise documentation. Optimized database schema by creating composite indexes on farmers and farms tables, verified foreign key integrity, and tested high-performance JOIN queries across districts and wards.
Friday *11/09/2026*	**INTERACTIVE GIS MAPPING & GEOJSON ORCHARD PLOTTING** Integrated GIS mapping module using Leaflet.js and OpenStreetMap API. Built backend endpoint to convert farm GPS coordinates into GeoJSON feature collections, enabling real-time interactive orchard pinpoints and ward centroid fallbacks.

DETAILS OF THE MAIN JOB OF THE WEEK

Operation: Enterprise Linux System Administration, Network Switch Troubleshooting, RBAC Security Implementation, and GIS Mapping Integration.

Machinery / Tools Used: Cisco Catalyst Switch, RJ45 Crimper & Cable Tester, Ubuntu Linux Terminal, XAMPP (Apache,

MariaDB), VS Code, Leaflet.js GIS Engine.

📸 WEEK 3 — PRACTICAL EVIDENCE & PHOTO GALLERY

Attach or paste printed field photographs, hardware inspections, and system screenshots in the designated boxes below:

[FIGURE 3.1: RJ45 T568B Patch Cord Crimping & Cable Testing]

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Technical Caption & Observations (07/09/2026):
Fabrication of Cat6 Ethernet patch cords according to T568B pinout standard and continuity verification with cable tester.

[FIGURE 3.2: Linux Server CLI & Process Monitoring]

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Technical Caption & Observations (08/09/2026):
Running Linux terminal administration commands (chmod, chown, ps aux, systemctl status apache2) in test environment.

[FIGURE 3.3: Leaflet.js GIS Interactive Farm Map]

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Technical Caption & Observations (11/09/2026):
Live browser view of interactive orchard map plotting avocado farms across Njombe wards with GeoJSON popups.

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WEEK NO: 4 | FROM: 14th September 2026 TO: 18th September 2026

DAY / DATE	ACTIVITY / TASKS EXECUTED
Monday *14/09/2026*	**PYTHON SCRIPTING FOR AUTOMATED DATABASE BACKUP & AUDIT** Developed Python automation script to connect with MySQL database. Automated scheduled database dumps, verified table row integrity, and generated automated system health summaries and export files for TTCL IT archives.
Tuesday *15/09/2026*	**RESTFUL API ENDPOINTS & ASYNCHRONOUS AJAX COMMUNICATION** Engineered RESTful JSON API endpoints for asynchronous client-server communication (api/get_wards.php). Implemented dynamic ward loading using JavaScript Fetch API without page reload, reducing server payload and improving frontend responsiveness.
Wednesday *16/09/2026*	**WEB APPLICATION SECURITY HARDENING: CSRF & XSS MITIGATION** Hardened web application security against cyber vulnerabilities. Implemented anti-CSRF cryptographic tokens (hash_equals) on all form submissions, enforced HTML output escaping against XSS, and secured user passwords using BCRYPT hashing algorithms.
Thursday *17/09/2026*	**MARKETPLACE ORDER DISPATCH & CONCURRENCY TRANSACTION LOCKING** Constructed buyer marketplace ordering system and order processing pipeline. Implemented database transaction locking (BEGIN...COMMIT) to prevent double-booking of harvest yields, and built receipt and delivery confirmation tracking module.
Friday *18/09/2026*	**DEVELOPER ENVIRONMENT SETUP, TESTING & SPRINT 2 DEMO** Configured developer environment (Node.js v24, Git 2.54, Python venv, IDE tools), conducted full end-to-end integration testing, and shared live preview link via Cloudflare tunnel. Successfully presented Sprint 2 milestone progress to supervisor.

DETAILS OF THE MAIN JOB OF THE WEEK

Operation: Python Script Automation, Full-Stack Web Security Hardening, Asynchronous REST API Development, and Marketplace Concurrency Testing.

Machinery / Tools Used: Python 3.11 Environment, Node.js, Git, Postman REST Client, Google Chrome DevTools, XAMPP (Apache, MySQL), Cloudflare Tunnel, VS Code.

📸 WEEK 4 — PRACTICAL EVIDENCE & PHOTO GALLERY

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[FIGURE 4.1: Python Database Automation Script Execution]

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Technical Caption & Observations (14/09/2026):
Running Python script with mysql-connector to perform automated backup dumps and data integrity checks.

[FIGURE 4.2: CSRF Protection & Security Audit in DevTools]

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Technical Caption & Observations (16/09/2026):
Inspecting HTTP POST requests with 64-character anti-CSRF tokens and XSS sanitized DOM nodes in Chrome DevTools.

[FIGURE 4.3: Live Cloudflare Tunnel Smartphone Demo & Sprint 2 Review]

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Technical Caption & Observations (18/09/2026):
Testing the Avocado Marketplace on mobile devices via encrypted Cloudflare Tunnel for supervisor demonstration.

COMPREHENSIVE MACHINERY, TOOLS USED & TECHNICAL OPERATIONS REPORT

This technical section provides a detailed engineering breakdown of all physical hardware, enterprise network machinery, operating system environments, development software, and industrial operations performed throughout the practical training tenure at Tanzania Telecommunication Corporation (TTCL).

Machinery / Tool Category	Technical Specifications	Operational Purpose in Field
Hardware Workstations & Peripherals	HP Compaq / ProDesk Desktop Workstations (Intel Core i5, 8GB/16GB DDR4 RAM, 500GB SSD/1TB HDD), HP LaserJet Enterprise Network Printers, UPS Back-UPS Units.	Daily host environment for systems development, preventative hardware servicing, dust purging, OS patches, and desktop disk cleanup.
Network Infrastructure Machinery	Cisco Catalyst 2960/3850 Managed Switches, Cat6 Unshielded Twisted Pair (UTP) Structured Cables, RJ45 Modular Plugs, Krone LSA Punch-down Tool, RJ45 Ratchet Crimping Tool, Master Cable Continuity Tester.	Local Area Network diagnostics, patch cord termination following TIA/EIA-568B standard, switch port link verification, and IP subnet communication testing.
Server & Operating System Platforms	Ubuntu Server 22.04 LTS (CLI), Windows 11 Pro Enterprise, Apache HTTP Server 2.4 (Daemon / Systemd service), MariaDB 10.4 / MySQL 8.0 Relational Database Server.	Linux CLI command administration, user permissions management (chmod/chown), daemon service monitoring (systemctl), and database transaction hosting.
Software Engineering & Development Tooling	Visual Studio Code / Antigravity AI IDE, Git Version Control (v2.54), Node.js Runtime (v24.16), Python 3.11 Environment (with mysql-connector, docx, reportlab), Postman API Client, Google Chrome DevTools.	Full-stack programming, REST API endpoint debugging, automated Python script creation for database backup, and asynchronous fetch testing.
GIS & Geolocation Mapping Engine	Leaflet.js 1.9+, OpenStreetMap Raster Tile Servers, GeoJSON FeatureCollection Specification, HTML5 Geolocation API.	Spatial coordinate plotting of smallholder avocado orchards, interactive map markers, popup card rendering, and ward centroid fallbacks.
Cybersecurity & Tunneling Infrastructure	BCRYPT Password Hashing Engine, Cryptographic Pseudo-Random Token Generator (bin2hex(random_bytes(32))), Cloudflare Named Tunnel (cloudflared), OpenSSL TLS 1.3.	Elimination of SQL Injection via prepared statements, anti-CSRF token verification, XSS sanitization, and secure public tunnel exposure for supervisor mobile evaluation.

DETAILED TECHNICAL OPERATIONS EXECUTED

Operation 1: Preventative Hardware Servicing & OS Performance Optimization

De-dusted workstation chassis using high-pressure blower. Inspected thermal paste contact between CPU and heatsink. Executed automated disk cleanup scripts, purged temporary files (%temp% and Prefetch), disabled unnecessary startup applications in Windows Task Manager, and verified RAM modules using Windows Memory Diagnostic. Result: Recovered 9.2 GB storage and significantly reduced operating system boot latency.

Operation 2: Structured Network Cabling & Managed Switch Diagnostics

Stripped Cat6 UTP cables and aligned 8 internal twisted conductors in strict adherence to TIA/EIA T568B pinout standard (White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown). Terminated RJ45 connectors using heavy-duty ratchet crimper. Verified end-to-end electrical continuity with cable tester. Conducted LAN ping diagnostics to gateway (192.168.1.1) and mapped active hosts using ARP cache tables.

Operation 3: Backend Architectural Refactoring & Database Security Layer

Eliminated legacy spaghetti database connection code by centralizing credentials in a secure .env file. Engineered config/db.php to enforce utf8mb4 multi-byte charset for seamless handling of Swahili diacritics. Suppressed display_errors in production mode and redirected unhandled database exceptions to logs/php_errors.log while presenting a graceful HTTP 500 error page. Refactored dynamic queries into parameterized prepared statements (mysqli_prepare), achieving 100% SQL injection immunity.

Operation 4: Agronomic Stock Lifecycle Engine & Concurrency Control

Modeled the avocado agricultural lifecycle via four phenological states: growing, flowering, harvest_ready, and harvested. Implemented automated tree count yield calculations in kilograms. Enforced farmer floor-price thresholds (min_price_per_kg) to protect smallholders from buyer market exploitation. Engineered database transaction locks (mysqli_begin_transaction ... FOR UPDATE) in market order submission to prevent double-booking of active harvest allocations.

Operation 5: Python Scripting for Database Automation & Automated Auditing

Engineered an automated Python script using mysql-connector and Python 3.11. The script establishes an encrypted local socket connection with the MariaDB database, verifies row counts across users and farms tables, cleans improperly formatted telephone records, and triggers scheduled mysqldump operations to produce dated .sql disaster recovery backup archives.

VIVA / ORAL DEFENSE CHEATSHEET & DAY 9–10 TECHNICAL NOTES

Essential Technical Explanations for Academic & Industrial Supervisors (Bilingual: English & Kiswahili)

Q1: Explain what you accomplished during 'Backend Refactoring and DB Connection' on September 3rd.

Technical Explanation (English): The initial prototype had database credentials hardcoded across various scripts, causing tight coupling and exposing sensitive server credentials whenever connection errors occurred. On September 3rd, I decoupled credentials into an environment file (.env), created an enterprise connection bootstrap (config/db.php) enforcing utf8mb4 encoding, silenced raw PHP errors in production to prevent information disclosure vulnerabilities, logged internal database exceptions to logs/php_errors.log, and refactored SQL queries to use parameterized prepared statements (mysqli_prepare).

Ufafanuzi kwa Kiswahili (Swahili Defense): Awali code ilikuwa na credentials za database ndani ya kila file (hardcoded) jambo ambalo lilikuwa hatari kiusalama. Tarehe 3 Septemba, nilitenganisha credentials kwenye faili la siri la .env, nikaweka encoding ya utf8mb4 kuzuia multibyte SQL injection na kusupport majina ya Kiswahili, nikaficha errors chafu kwa wateja kwa kuweka custom 500 error page, na kuandika makosa yote kisiri kwenye logs/php_errors.log, pamoja na kuweka msingi wa prepared statements kuzuia SQL injection kwa 100%.

Q2: How does the 'Product Inventory and Stock Module' (September 4th) benefit smallholder farmers?

Technical Explanation (English): The module transforms the system from a static registry into a dynamic agricultural management system. It tracks the crop phenological lifecycle through four distinct states: growing, flowering, harvest-ready, and harvested. It calculates expected yield in kilograms based on tree count metrics. Crucially, it enforces a farmer floor-price protection threshold (min_price_per_kg) below which buyers cannot bid, and utilizes database transaction locks to reserve stock atomically when orders are placed.

Ufafanuzi kwa Kiswahili (Swahili Defense): Moduli hii inamsaidia mkulima wa parachichi kwanza kwa kufuatilia mzunguko halisi wa zao shambani (growing, flowering, na harvest-ready) ili mnunuzi aone lini mzigo utakomaa. Pili, inampa ulinzi wa bei (Floor Price) kupitia min_price_per_kg ambapo hakuna dalali anayeweza kuagiza mzigo chini ya bei hiyo. Tatu, inalinda mzigo usinunuliwe mara mbili kwa kutumia database transaction locking pindi mnunuzi anapoweka oda.

Q3: Why did you develop the system using native PHP instead of a framework like Laravel?

Technical Explanation (English): Native PHP 8.2 and Vanilla JavaScript provide zero-dependency, plug-and-play hosting. The system can be deployed immediately on municipal or TTCL local servers without requiring Composer, Node.js build processes, or extensive memory overhead. It ensures optimal execution speed and gives full transparency over application security mechanisms.

Ufafanuzi kwa Kiswahili (Swahili Defense): Kutumia native PHP kunaiwezesha system kufanya kazi mara moja (plug-and-play) kwenye server yoyote ya TTCL au halmashauri bila kuhitaji kusanidi Composer au Node build tools, na inatoa kasi kubwa ya utendaji bila kutumia RAM kubwa.

Q4: How did you implement cybersecurity safeguards (SQL Injection, CSRF, and XSS)?

Technical Explanation (English): SQL Injection was eliminated through 140+ parameterized prepared statement calls (mysqli_prepare). CSRF is mitigated by attaching a 64-character cryptographically secure token (bin2hex(random_bytes(32))) to every session and validating it with hash_equals() upon POST submission. XSS is prevented by passing all dynamic database outputs through htmlspecialchars() before rendering in the DOM.

Ufafanuzi kwa Kiswahili (Swahili Defense): Tulitumia Prepared Statements kwa 100% kuzuia SQL Injection ili data ya mtumiaji isichanganywe na SQL. Kwenye fomu zote tuliweka CSRF token ya siri inayothibitishwa kwa hash_equals(), na data zote zinazoonyeshwa kwenye skrini zinasafishwa kwa htmlspecialchars() kuzuia XSS attacks.

Q5: What was the role of Python in this project when the main system is built in PHP?

Technical Explanation (English): Python served as an enterprise backend automation and data auditing utility. I developed scripts to establish database connections, automate scheduled database backup dumps (.sql), verify smallholder contact data integrity, and generate executive summaries for IT archive reporting without manual phpMyAdmin intervention.

Ufafanuzi kwa Kiswahili (Swahili Defense): Python ilitumika kama chombo cha otomatiki (Automation Tool). Nilijenga script ya Python kufanya backup za kiotomatiki za database ya MySQL, kusafisha data za wakulima (Data Cleansing), na kutoa ripoti za kiutawala kiotomatiki.